

SEUPD@CLEF: Team NEXA

Paul Arlot¹, Andrea Di Tillo¹, Gaute Greiff Flagstad¹, Bitu Khashechian¹,
Danil Smirnov¹ and Marco Tomaioli¹

¹University of Padua, Italy

Abstract

This paper presents the system developed by Team NEXA for Task 1 of the CLEF 2026 CheckThat! Lab, focusing on the retrieval of scientific sources for claims found on social media. Given the challenge of linking informal web discourse to formal scientific literature across English, German, and French, we propose a multi-stage retrieval framework. Our approach combines traditional lexical search using BM25 with a neural re-ranking component based on multilingual transformer models. The system aims to bridge the linguistic and stylistic gap between social media posts and academic abstracts, providing a scalable solution to assist fact-checkers in verifying scientific information.

Keywords

Information Retrieval, Search Engine, CLEF, CheckThat!, Scientific Claim Verification, Lucene, Multilingual Search

1. Introduction

Information Retrieval systems play a fundamental role in enabling access to relevant data across massive, often heterogeneous corpora. In the context of the CheckThat! Lab at CLEF, which focuses on identifying the original scientific research behind social media claims, we have developed a configurable IR system. The motivation for this project comes from the need to fight disinformation by making fact-checking faster. As noted by CheckThat!, manually searching for scientific evidence is very slow for human experts. By automating this source retrieval process, we can help fact-checkers quickly link a claim to its original paper. A primary difficulty in this task is cross-language retrieval: social media posts formulated in English, French and German must be mapped to a corpus of scientific publications that is mainly in English, alongside a minority of other languages. To achieve this, we translated every post and document in English in order to be able to use the same approach regardless of the source language. Our system supports a wide range of preprocessing and indexing strategies, enabling us to experiment with different setups to address the vocabulary gap between informal web text and formal scientific abstracts.

The paper is organized as follows: Section 2 describes our approach; Section 3 explains our experimental setup; Section 4 discusses our main findings; finally, Section 5 draws some conclusions and outlooks for future work.

2. Methodology

In this section, we describe the architecture and components of our system. Our approach follows a multi-stage Information Retrieval pipeline combining traditional lexical matching with semantic representations.

“Search Engines”, course at the master degree in “Computer Engineering”, Department of Information Engineering, and at the master degree in “Data Science”, Department of Mathematics “Tullio Levi-Civita”, University of Padua, Italy. Academic Year 2025/2026

EMAIL: paulouisjean.arlot@studenti.unipd.it (P. Arlot); andrea.ditillo@studenti.unipd.it (A. D. Tillo); gaute.greiff.flaegstad@studenti.unipd.it (G. G. Flagstad); bitu.khashechian@studenti.unipd.it (B. Khashechian); danil.smirnov@studenti.unipd.it (D. Smirnov); marco.tomaioli@studenti.unipd.it (M. Tomaioli)



© 2026 Copyright for this paper by its authors.

Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

2.1. System Configurability

2.2. Data Parsing

The first stage of the pipeline turns the two textual inputs of the system, the corpus of scientific publications and the set of input claims, into structured Java objects that the rest of the pipeline can work with. Both inputs are provided as JSON and share the same text cleaning routine, so we describe them together. On the data side we have two simple classes: `Publication` for documents and `Claim` for queries.

2.2.1. Documents (Publications)

Documents are read through an abstract `CommonParser` class that defines the iterator interface and exposes the shared text cleaning routine, with concrete subclasses for each input format. Since the corpus is provided as JSON, the only subclass we currently use is `JsonParser`.

`CommonParser` implements `Iterator<Publication>`, so it produces one `Publication` at a time as the input is read. Each `Publication` gets the fields we use for retrieval: a numerical identifier (pubkey), the title, abstract, venue and authors. Missing or null fields are replaced with empty defaults, and empty abstracts are filled with a placeholder token so that every document still produces a valid entry in the index.

`JsonParser` uses the Jackson library to read the corpus, which is stored as a JSON array of document objects. We process the stream incrementally rather than loading the whole file at once, which keeps memory usage flat regardless of corpus size. For each publication we map the JSON fields onto a `Publication` instance and run the abstract through the cleaning routine before it leaves the parser.

2.2.2. Queries (Claims)

Queries are loaded from a separate JSON topics file containing a list of claim objects. The file is small and only read once at the start of a run, so we parse it into a `List<Claim>` using a Jackson `ObjectMapper`, rather than going through the streaming parser used for documents. Each `Claim` keeps all three fields: the claim index (used as the query identifier in the TREC run file), the claim text, and the gold pubkey when present (used only at evaluation time).

To keep documents and queries aligned, the cleaning routine described next is applied to the claim text from inside the setter of the `Claim` class. By the time the searcher iterates over the list, both sides of the retrieval problem have gone through exactly the same preprocessing, which avoids the kind of subtle mismatches that arise when documents and queries are normalised differently.

2.2.3. Text Cleaning

Before any text reaches the analyzer, we pass it through a cleaning function (`cleanText`) that strips out the noise produced by the mix of scientific writing and social media content in the corpus. The function applies the following steps in order:

- **Unicode normalisation:** the text is converted to NFC form so the character representation is consistent.
- **HTML unescaping and stripping:** HTML entities are decoded and any leftover tags are removed.
- **URL removal:** hyperlinks are dropped so they do not end up as tokens.
- **Citation removal:** bracketed references (e.g. “[1, 2–5]”) and author–year citations (e.g. “(Rossi et al., 2023)”) are removed.
- **Section header removal:** structural labels that appear in scientific abstracts, such as *Abstract*, *Methods*, *Results* or *Conclusions*, are stripped.

- **Symbol and emoji removal:** mathematical and scientific symbols (e.g. $^{\circ}$, $\pm$, $\geq$) and emoji characters are replaced with spaces.
- **Social media artefacts:** Twitter-style mentions (e.g. @user) are removed.
- **Whitespace collapsing:** runs of whitespace are collapsed into a single space and any leading or trailing whitespace is trimmed.

This step is important because the corpus mixes formal abstracts with informal claims. Without it the analyzer would have to deal with a lot of noisy tokens that could decrease matching quality.

2.3. Query Expansion

2.4. Text Analysis

The text analysis phase transforms raw textual input into a stream of tokens that can be indexed and later retrieved. Initially, we tried to use language-specific analyzer classes but using a unique English Analyser after translation was simpler and offered better results. The analysis is performed through an analyzer classes, which extend the Apache Lucene **Analyzer** framework.

The analyzer follows an architecture composed of three main components:

- **Tokenizer:** Responsible for splitting the input text into raw tokens.
- **Token Filters:** Sequentially modify, normalize, or enrich the token stream.
- **Stemming/Lemmatization:** Reduces tokens to their base or root forms to improve generalization.

The overall pipeline is dynamically configured through external YAML configuration files, allowing flexible adaptation of processing strategies and experimental settings.

2.4.1. Tokenizer Configurations

Multiple tokenization strategies are supported, enabling the system to handle different types of input text:

- **Standard Tokenizer:** Uses Lucene's rule-based tokenizer to handle punctuation, digits, and word boundaries.
- **Letter Tokenizer:** Emits tokens composed only of alphabetic characters, treating non-letter characters as delimiters.
- **Whitespace Tokenizer:** Splits input strictly on whitespace characters without processing punctuation.
- **NLP Tokenizer:** Uses Apache OpenNLP models to perform linguistically informed tokenization.

The tokenizer type is selected through configuration files, allowing to try different strategies.

2.4.2. Token Filters

Token filters are applied after tokenization to normalize and enrich the token stream. These filters are categorized into general (language-independent) and language-specific filters.

General Token Filters These filters perform general normalization operations across all languages:

- **LowerCaseFilter:** Converts all tokens to lowercase.
- **ICUFoldingFilter:** Normalizes accented and special characters into ASCII equivalents.

- **NBSPFilter:** Removes non-breaking spaces and trims tokens.
- **RemoveDuplicatesTokenFilter:** Eliminates duplicate tokens.
- **LengthFilter:** Removes tokens that are too short or too long.
- **RepeatedLetterFilter:** Normalizes tokens with repeated characters (e.g., “soooo” → “soo”).

Enrichment Filters To improve semantic representation, additional filters are applied:

- **AbbreviationExpansionFilter:** Expands abbreviations using predefined mappings (e.g., “ml” → “machine learning”).
- **CompoundPOSTokenFilter:** Uses Part-of-Speech tagging to combine semantically related tokens into compound expressions.
- **ShingleFilter:** Generates n-grams to capture local contextual information.
- **PositionFilter:** Normalizes positional increments for indexing consistency.

Language-Specific Filters Each analyzer applies additional filters tailored to linguistic characteristics:

- English-specific filters (e.g., possessive removal, English stopwords)
- French-specific filters (e.g., elision handling, French stopwords)
- German-specific filters (e.g., compound handling and normalization)

This design ensures that each language is processed according to its grammatical and lexical properties. Since we translate everything to English, we are using only English-specific filters.

2.4.3. Stemming and Lemmatization

The system supports multiple strategies for reducing tokens to their canonical forms: **Porter Stemmer**, **Snowball Stemmer**, **OpenNLP Lemmatizer**, **No Stemming**.

The selected method is controlled through configuration files and can vary depending on the experimental setup.

2.5. Document Indexing

The document indexing stage transforms processed publications into a searchable inverted index using Apache Lucene. This structure enables efficient retrieval of relevant documents during the query phase. In our system, indexing is handled by the **DirectoryIndexer** component, which coordinates parsing, preprocessing, and storage of documents.

2.5.1. Indexing Workflow

The indexing workflow operates over a collection of JSON files containing scientific publications. Each file is parsed to extract individual documents, represented internally as structured **Publication** objects. Only documents containing meaningful textual content (such as title or abstract) are considered for indexing.

In addition to standard preprocessing, the system supports optional enhancements. When enabled, documents written in languages different from the target language can be translated prior to indexing. Moreover, the system can generate dense vector representations of the textual content by interacting with an external embedding service. These embeddings are stored alongside the indexed documents to support future semantic retrieval extensions.

Each processed publication is converted into a Lucene **Document** and added to the index using an **IndexWriter**. During this process, statistics such as the number of indexed documents and total indexing time are recorded.

2.5.2. Field Representation

Document content is stored in structured fields, such as title and abstract. A custom field type is defined to support advanced indexing features, including term frequencies, positional information, and character offsets.

The system also enables storage of term vectors, which allows more advanced retrieval functionalities such as phrase matching, proximity queries, and potential re-ranking strategies. Additionally, the original textual content can be stored in the index to support downstream processing steps.

2.5.3. Processing Pipeline

The indexing procedure follows a well-defined sequence of steps. First, the system initializes the index writer with the configured analyzer and output directory. Then, for each document, the following operations are performed:

- Parsing the input file into structured publication objects
- Filtering out incomplete or invalid entries
- Detecting the language of the document
- Applying optional translation if required
- Generating semantic embeddings (if enabled)
- Converting the document into a Lucene-compatible format
- Adding the document to the index

Once all documents have been processed, the index is finalized by committing changes and closing the writer.

This modular and extensible design allows the indexing system to integrate multiple preprocessing strategies, multilingual handling, and semantic enrichment, while keeping the core indexing logic independent and reusable.

2.6. Retrieval Phase

The retrieval phase is responsible for matching input claims with relevant scientific publications stored in the indexed collection. This component is implemented through the **Searcher** class, which performs query processing, executes search operations over the Lucene index, and outputs ranked results.

The system initializes a Lucene **IndexSearcher** over the indexed data and employs the BM25 similarity function for ranking. Queries are provided as a collection of claims stored in JSON format, where each claim represents a short textual statement that must be linked to relevant documents.

2.6.1. Query Formulation

Each claim is processed independently. The textual content is first transformed into a structured query using a **QueryBuilder**, which applies the same analyzer used during indexing. This ensures that both documents and queries undergo consistent preprocessing, reducing mismatches due to tokenization or normalization.

The resulting terms are used to construct field-specific queries targeting different parts of the indexed documents.

2.6.2. Structured Boolean Queries

Instead of relying on a single-field representation, the system generates separate subqueries for the title and abstract fields of each document.

These subqueries are combined using a Boolean query with **SHOULD** clauses, allowing documents to be retrieved if they match either field. This formulation increases recall while maintaining flexibility in matching.

To further refine the ranking, different importance weights are assigned to each field through boosting. In particular, matches occurring in the title field are given higher importance than those in the abstract. This results in a weighted query structure that prioritizes concise and informative matches while still considering broader contextual information.

2.6.3. Ranking Mechanism

Once the Boolean query is constructed, it is executed using the Lucene search engine. The system retrieves the top- k documents according to their BM25 scores. This ranking function considers the frequency of query terms, their rarity across the collection, and document length normalization.

The number of retrieved documents is configurable and can be adjusted depending on the evaluation setup.

2.6.4. Result Formatting

The retrieved results are written to an output file following the TREC evaluation format. Each entry includes the query identifier, document identifier, rank position, score, and run identifier. This standardized format ensures compatibility with external evaluation tools commonly used in information retrieval benchmarks.

3. Experimental Setup

Describe the experimental setup, i.e.

- used collections
- evaluation measures
- url to git repository and its organization
- hardware used for experiments
- ...

4. Results and Discussion

In this section we report the experiments that led to the final lexical and hybrid configurations of the system. Because the search space is much larger than what we could grid-search, we tuned the components in order, fixing the winner of each phase before moving on:

1. Section 4.1 fixes the analyzer pipeline (tokenizer, stemmer, stop list, filters).
2. Section 4.2 fixes the retrieval model and BM25 parameters.
3. Section 4.3 fixes the fielded query (boosts, query mode, IDF pruning).
4. Section 4.4 reports two expansion strategies (WordNet, PRF) that did not work.
5. Section 4.5 ablates the components of the final lexical system.
6. Section 4.6 reports the dense fusion and reranking results.
7. Section 4.7 compares the runs we submitted.

All scores are MRR@5 on the dev split, reported per language (EN, FR, DE) with the average across languages. Hybrid and reranking experiments operate on the top-50 candidates of the lexical run. The dataset, dev splits, hardware, and evaluation tooling are described in Section 3.

4.1. Lexical Analyzer Tuning

We start by fixing the analyzer pipeline that the rest of the chapter builds on: the tokenizer, the stop list, the stemming strategy, and a few token filters.

Table 1

Tokenizer and stemmer comparison on the EN/FR/DE dev sets, MRR@5. Variants share the standard pipeline (lowercase, ICU folding, English stop list) and differ only in the stemmer or tokenizer.

Configuration	EN	FR	DE	Avg
Standard, no stop, no stem	0.4826	0.4453	0.3014	0.4098
Standard, stop, no stem	0.4864	0.4674	0.3225	0.4254
Standard, stop, EnglishMinimal	0.4894	0.4817	0.3304	0.4338
Standard, stop, Snowball	0.4869	0.4785	0.3358	0.4337
Standard, stop, Porter	0.4867	0.4801	0.3381	0.4350
Standard, stop, KStem	0.4883	0.4901	0.3354	0.4379
Whitespace, stop, KStem	0.4422	0.4087	0.2746	0.3752

4.2. Retrieval Model and BM25 Parameters

With the analyzer fixed, we choose a similarity function and tune its parameters. We compare BM25 against Lucene’s other built-in similarity models, then sweep BM25’s k_1 and b over a small grid.

Table 2

Retrieval model comparison, MRR@5 on the dev sets. BM25 uses Lucene defaults ($k_1=1.2$, $b=0.75$); only the most informative LM Dirichlet and LM Jelinek-Mercer rows are shown.

Model	EN	FR	DE	Avg
BM25 ($k_1=1.2$, $b=0.75$)	0.4838	0.4887	0.3342	0.4356
Classic (TF-IDF)	0.4130	0.3948	0.2715	0.3597
LM Dirichlet ($\mu=500$)	0.4683	0.4652	0.3357	0.4231
LM Dirichlet ($\mu=1000$)	0.4526	0.4434	0.3098	0.4019
LM Jelinek-Mercer ($\lambda=0.5$)	0.4784	0.4706	0.3220	0.4236

4.3. Fielded Search and Query Construction

We now tune how the query is assembled: which fields to boost, whether terms are required or optional, and how aggressively to prune the query.

4.4. Failed Expansion Approaches

We tested two query expansion strategies that the methodology supports, WordNet synonyms and pseudo-relevance feedback. Both hurt retrieval, and for the same underlying reason.

4.5. Ablation of the Lexical System

To isolate the contribution of each lexical component, we ablate them one at a time on top of the full best configuration, sharing a single index across runs.

Table 3

Field boost sweeps on the EN/FR/DE dev sets, MRR@5. Part A varies the title boost with abstract fixed at 1.0; Part B varies the abstract boost with title fixed at 2.0; Part C adds the authors boost on top of title² abstract³.

Configuration	EN	FR	DE	Avg
<i>Part A: title boost (abstract¹)</i>				
title ^{1.0}	0.4997	0.5026	0.3577	0.4533
title ^{1.5}	0.4929	0.4964	0.3475	0.4456
title ^{2.0}	0.4838	0.4882	0.3352	0.4357
title ^{3.0}	0.4635	0.4567	0.3159	0.4121
<i>Part B: abstract boost (title²)</i>				
abstract ^{1.0}	0.4838	0.4882	0.3352	0.4357
abstract ^{1.5}	0.4956	0.5000	0.3506	0.4487
abstract ^{2.0}	0.4997	0.5026	0.3577	0.4533
abstract ^{3.0}	0.4990	0.5066	0.3604	0.4553
<i>Part C: authors boost (title² abstract³, topK=30)</i>				
authors ^{0.5}	0.4888	0.4938	0.3373	0.4400
authors ^{1.0}	0.4990	0.5066	0.3604	0.4693
authors^{1.5}	0.5172	0.5145	0.3841	0.4732
authors ^{2.0}	0.5160	0.5130	0.3820	0.4715
authors ^{3.0}	0.5050	0.5020	0.3754	0.4608

Table 4

IDF-based query pruning on top of the fielded query, MRR@5. Each claim is tokenised with the English analyzer; only the top- K tokens by IDF are passed as the query.

K	EN	FR	DE	Avg
all (no pruning)	0.4838	0.4882	0.3352	0.4357
10	0.4115	0.4189	0.2887	0.3730
15	0.4696	0.4658	0.3182	0.4179
20	0.4914	0.4798	0.3379	0.4364
30	0.4956	0.4860	0.3395	0.4404
40	0.4956	0.4860	0.3395	0.4404
50	0.4956	0.4860	0.3395	0.4404

Table 5

Query expansion experiments, MRR@5. All runs use the lexical configuration fixed in Sections 4.1–4.3.

Configuration	EN	FR	DE	Avg
Lexical baseline (no expansion)	0.4883	0.4901	0.3354	0.4379
WordNet, max 1 syn/word	0.4582	0.4352	0.3040	0.3991
WordNet, max 3 syn/word	0.4262	0.3993	0.2829	0.3695
WordNet, max 5 syn/word	0.4047	0.3759	0.2591	0.3466
PRF (3 docs, 3 terms)	0.4825	0.4763	0.3351	0.4313
PRF (5 docs, 3 terms)	0.4829	0.4790	0.3310	0.4310
PRF (10 docs, 10 terms)	0.4776	0.4692	0.3317	0.4262

4.6. Hybrid Retrieval: Lexical and Dense Fusion

On top of the lexical system fixed in the previous sections, we add a dense retrieval signal from a multilingual sentence encoder and combine the two scores. We compare two encoder families and two ways of combining the scores.

Table 6

Ablation of the lexical system, MRR@5. The reference is the full configuration (BM25 $k_1=1.0$, $b=0.75$, title² abstract³ authors^{1,5}, topK=30). Each row removes or changes exactly one component.

Configuration	EN	FR	DE	Avg	Δ Avg
Full best (reference)	0.5172	0.5145	0.3841	0.4719	—
– authorsBoost (set to 0)	0.5086	0.4973	0.3646	0.4568	−0.0151
– topKQueryTerms (use all terms)	0.5100	0.5131	0.3826	0.4686	−0.0034
– abstractBoost (restore abs ¹)	0.4983	0.4885	0.3548	0.4472	−0.0247
– all three (simple baseline)	0.4874	0.4919	0.3323	0.4372	−0.0347
+ venueBoost = 0.5	0.5174	0.5155	0.3835	0.4721	+0.0002
queryMode = mixed (must=2)	0.0752	0.1233	0.0635	0.0873	−0.3846
queryMode = must (all)	0.0151	0.0043	0.0026	0.0073	−0.4646

4.6.1. BGE-M3 Reranking and Fusion

We first compare two ways of using BGE-M3 embeddings: pure reranking, which reorders the top-50 lexical hits by cosine similarity, and score fusion, which weights the BM25 and cosine scores against each other within each query.

Table 7

Lexical baseline vs. BGE-M3 reranking and fusion, MRR@5. Reranking reorders the top 50 lexical hits by cosine; fusion combines $(1-\alpha) \cdot \text{BM25}_{\text{norm}} + \alpha \cdot \text{cosine}$ with both scores min-max normalised within each query's top-50.

Method	EN	FR	DE	Avg
Lexical baseline	0.5172	0.5145	0.3841	0.4719
BGE-M3 reranking only	0.5256	0.5280	0.3967	0.4834
BGE-M3 fusion ($\alpha=0.70$)	0.5768	0.5770	0.4331	0.5290

4.6.2. SPECTER2 vs BGE-M3

BGE-M3 is a general-purpose multilingual encoder. We also tested SPECTER2, a domain-specific encoder trained on scientific paper-to-paper citation proximity, to see whether the scientific specialisation outweighs the loss of multilingual coverage. It does not.

Table 8

SPECTER2 vs. BGE-M3 at their respective best fusion weights. Both use the same 50-candidate input lists from the lexical run.

Method	EN	FR	DE	Avg
Lexical baseline	0.5172	0.5145	0.3841	0.4719
SPECTER2 reranking only	0.4451	0.4233	0.3377	0.4020
SPECTER2 fusion ($\alpha=0.60$)	0.5524	0.5475	0.4150	0.5050
BGE-M3 fusion ($\alpha=0.70$)	0.5768	0.5770	0.4331	0.5290

4.7. Final Submitted Runs

Combining the winners of each phase above gives the configurations we submit to the official evaluation.

Table 9

Submitted runs and dev-set MRR@5 per language.

Run	Description	EN	FR	DE	Avg
Run 1	Lexical only (best lexical config)	0.5172	0.5145	0.3841	0.4719
Run 2	Hybrid: lexical + BGE-M3 fusion ($\alpha=0.70$)	0.5768	0.5770	0.4331	0.5290

5. Conclusions and Future Work

Provide a summary of what are the main achievements and findings.

Discuss future work, e.g. what you may try next and/or how your approach could be further developed.

6. Group Members Contribution

Jane Doe Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

John Doe Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Jane Doe Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

John Doe Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Jane Doe Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

John Doe Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor



Figure 1: 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (<https://goo.gl/VLCRBB>).

Table 10
Frequency of Special Characters

Non-English or Math	Frequency	Comments
∅	1 in 1,000	For Swedish names
π	1 in 5	Common in math
\$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

7. Misc [TO BE REMOVED]

7.1. Tex Files

Put your L^AT_EX files into the `section` folder as shown in the examples above.

7.2. Figures

Put your figures into the `figure` folder and put the caption under the figure. Example of reference to Figure 1.

7.3. Tables

Put the caption above the table. Example of reference to Table 10.

See the `booktab` packaged documentation for further options.

7.4. Bibliography

Example of citations:

- name: ?]
- parenthesis: [?]

An initial list of references is provided in the files `bibliography.bib` and `proceedings.bib` that you can expand.

See the `natbib` packaged documentation for further options.

7.5. Acronyms

Use the

`\ac{acronym}`

command to insert acronyms, eg. *Average Precision (AP)*. The command will expand the acronym the first time it is used.

An initial list of acronyms is provided in the file `acronyms.tex` that you can expand.

See the `acronym` packaged documentation for further options.